

MATH 3430 - Week 3

The Laplace Transform

1 Overview

PLEASE NOTE: We are still going to work through some material that is contained in the Week 2 Notes for the first few days this week. Refer to those notes for the material on Planar Systems. These notes just contain material related to the Laplace Transform. We will (hopefully) start The Laplace Transform on Wednesday.

This week we introduce and explore the **Laplace Transform**. This is an extremely powerful tool for solving **constant coefficient nonhomogeneous IVPs**. It works by transforming differential equations (in the variable t) into algebraic equations (in the variable s). If we can manipulate/solve the algebraic equation as a function of s , then we can transform back to a solution to the differential equation (as a function of t).

Since we already have seen some non-transform techniques for solving nonhomogeneous IVPs when the forcing terms are made up of “elementary” functions (polynomials, exponentials, and trigonometric functions), the emphasis this week will be on forcing terms that are “non-elementary.” This will focus on non-continuous forcing terms, mostly made up of piece-wise defined functions. A special case of this are **step functions**, most importantly the **unit step function**. We will also look at a very important example of what are called **generalized functions** (also called distributions). The most important of these is the **Dirac Delta Function** (also called the **unit impulse function**).

Given a function $f(t)$ (which satisfies certain conditions that we won’t go into here), we define the Laplace Transform of f , written $L[f](s)$ by the following:

$$L[f](s) = \int_0^{\infty} e^{-st} f(t) dt$$

This is called an **integral transform** since we are integrating the given function $f(t)$ to get a new function $L[f](s)$.

A very important idea that we will explore is that of the **convolution** of two functions. The convolution of f and g is written $f * g$ and is defined in the following way:

$$(f * g)(t) = \int_0^t f(t-x)g(x)dx$$

This will allow us to understand how the products of two Laplace Transforms relates to the original functions themselves.

While we won’t go very far in our exploration in only a week, the ideas and methods of this material extends to a whole range of other topics in Analysis. These include Fourier Transforms, Integral Transforms in general, Partial Differential Equations, Complex Analysis, Harmonic Analysis, and Functional Analysis. Thus, the Laplace Transform is an introduction to almost all of the main ideas of Classical Analysis.

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) Distinct Real Eigenvalues. Classification of Equilibrium Points.
- (T) Complex Eigenvalues. Repeated Real Eigenvalues. The Trace-Determinant Plane.
- (W) Definition and Basic Properties of The Laplace Transform. Solving IVPs. Partial Fractions.
- (R) Electrical Circuits. Discontinuous Forcing Functions. The Unit Step and Impulse (Dirac Delta) Function.
- (F) Convolution. (More) Advanced Properties.

Here is a checklist of things to do this week. This is meant to help you keep track of what you've completed and what you still need to work on.

- ☐ Read these notes and ask any questions that you have about anything.
- ☐ Read the sections from Judson and Lebl.
- ☐ Complete all of the homework exercises.

3 Terminology and Notation

- The Laplace Transform

$$L[f](s) = F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

This is defined if the improper integral converges. If $f(t), g(t)$ are functions with Laplace Transforms $F(s), G(s)$, then

$$L[\alpha f + \beta g](s) = \alpha L[f](s) + \beta L[g](s) = \alpha F(s) + \beta G(s)$$

This says that the Laplace Transform is a linear operator (an operator is a fancy name for something that maps functions to functions, in this case $L : f \rightarrow F$).

- The Inverse Laplace Transform

The Inverse Laplace Transform, L^{-1} , is an operator that maps $F(s)$ to $f(t)$: $L^{-1} : F \rightarrow f$. It can be defined in terms of integrals, but it involves complex numbers. You do not need to know the integral definition.

The Inverse Laplace Transform is also linear:

$$L^{-1}[\alpha F + \beta G](t) = \alpha L^{-1}[F](t) + \beta L^{-1}[G](t) = \alpha f(t) + \beta g(t)$$

- Exponentially Bounded Functions: $f(t)$ is exponentially bounded if there exist constants $M > 0$ and a such that

$$|f(t)| \leq Me^{at}$$

If $f(t)$ is exponentially bounded, then the Laplace Transform of f exists.

- Transforms of IVP: let $y(t)$ be a piecewise continuous exponentially bounded function with Laplace Transform $Y(s)$. If $y'(t)$ is also exponentially bounded, then the Laplace Transform of $y'(t)$ exists and

$$L[y'](s) = sL[y](s) - y(0) = sY(s) - y(0)$$

If $y''(t)$ is also exponentially bounded, then

$$L[y''](s) = s^2L[y](s) - sy(0) - y'(0) = s^2Y(s) - sy(0) - y'(0)$$

Similar expressions exist for all $y^{(n)}(t)$. All of these can be proved using integration by parts the appropriate number of times.

The important thing to take away from all of this is that the Laplace Transform sends derivatives (of y) to polynomials (of s) and the initial conditions show up as the coefficients of the polynomials.

Let $y(t)$ and $f(t)$ be an exponentially bounded functions with Laplace Transforms $Y(s)$ and $F(s)$ and let $y'(t)$ and $y''(t)$ be exponentially bounded. Combining the above results with the linearity of the Laplace Transform, the IVP

$$ay''(t) + by'(t) + cy(t) = f(t), \quad y(0) = y_0, \quad y'(0) = y'_0$$

becomes the algebraic equation

$$a[s^2Y - sy_0 - y'_0] + b[sY - y_0] + cY = F$$

which can be solved for $Y(s)$:

$$Y(s) = \frac{F(s) + a(sy_0 + y'_0) + by_0}{as^2 + bs + c}$$

The polynomial in the denominator is the **characteristic polynomial** of the second-order homogeneous differential equation $ay'' + by' + c = 0$.

4 Equations and Models

- Electrical Circuits

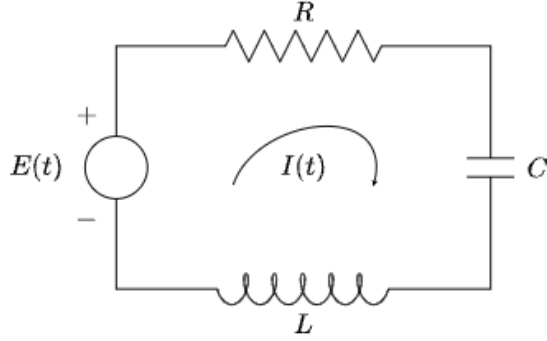


Figure 1: A simple RLC circuit

The diagram shows the schematic for a simple **RLC Circuit**. In any circuit, the main objects under consideration are the **charge** on the capacitor at time t , which is denoted by $Q(t)$ and the **current** through the capacitor, which is denoted by $I(t)$. The relationship between the two is given by $I(t) = Q'(t)$. The function $E(t)$ is the voltage (from a battery or some other external source), and the constants, R, C, L are determined by the components of the circuit. R is the resistance, C is the capacitance, and L is the inductance.

Using various physical laws (Faraday's Law and Kirchoff's Laws) we have the following equivalent second-order differential equations

$$LQ'' + RQ' + \frac{1}{C}Q = E(t), \quad \text{and} \quad LI'' + RI' + \frac{1}{C}I = E'(t)$$

5 Important Formulas (analytical material)

- The unit step $u_c(t)$ function is given by

$$u_c(t) = \begin{cases} 0 & \text{if } t < c \\ 1 & \text{if } t \geq c \end{cases}$$

The Laplace Transform of $u_c(t)$ is

$$L[u_c(t)](s) = \frac{e^{-cs}}{s}$$

- The **Dirac delta function**, $\delta(t)$, is the **generalized function** defined by $\delta(t) = 0$ if $t \neq 0$ and

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

The **unit impulse function**, $\delta_{t_0}(t)$ can be defined as a shifted Dirac delta function: $\delta_{t_0}(t) = \delta(t - t_0)$.

Impulse functions are only properly defined inside of definite integrals. They have the property that if $f(t)$ is defined at t_0 then

$$\int_{-\infty}^{\infty} \delta_{t_0}(t) f(t) dt = f(t_0)$$

The Laplace Transform of $\delta_{t_0}(t)$ is given by

$$L[\delta_{t_0}](s) = e^{-t_0 s}$$

- Given two functions $f(t)$ and $g(t)$, the convolution of f and g , denoted $(f * g)(t)$, is given by

$$(f * g)(t) = \int_0^t f(t - \tau)g(\tau)d\tau$$

6 Geometric and Visual Aspects (qualitative material)

- This is the **unit step function**, $u_c(t)$ with $c = 0$.

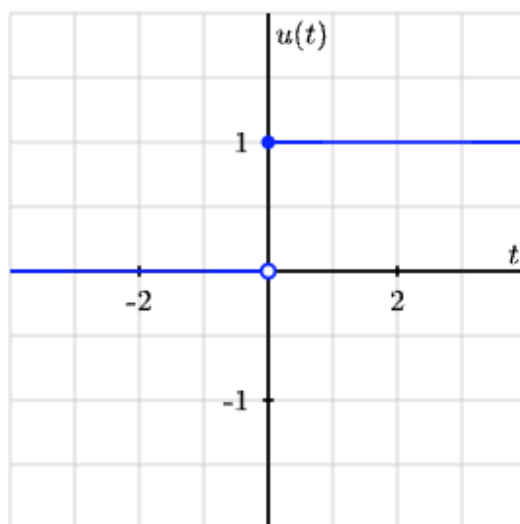


Figure 2: Plot of the Unit Step function $u_0(t)$.

- Table of Laplace Transform Pairs: $f(t)$ and $F(s)$

$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$
e^{at}	$\frac{1}{s-a}$
$t^n, \quad n = 1, 2, 3, \dots$	$\frac{n!}{s^{n+1}}$
$t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$
$t^{n-\frac{1}{2}}, \quad n = 1, 2, 3, \dots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1) \sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
$\sin(at)$	$\frac{a}{s^2 + a^2}$
$\cos(at)$	$\frac{s}{s^2 + a^2}$
$f(ct)$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
$u_c(t) = u(t-c)$ Heaviside Function	$\frac{e^{-cs}}{s}$
$\delta(t-c)$ Dirac Delta Function	e^{-cs}
$u_c(t) f(t-c)$	$e^{-cs} F(s)$
$u_c(t) g(t)$	$e^{-cs} \mathcal{L}\{g(t+c)\}$
$e^{ct} f(t)$	$F(s-c)$
$t^n f(t), \quad n = 1, 2, 3, \dots$	$(-1)^n F^{(n)}(s)$
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2 F(s) - sf(0) - f'(0)$
$f^{(n)}(t)$	$s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0)$

7 Sage Code (numerical material)

There is brief introduction to using Sage to find Laplace Transforms in section 6.1.8 of Judson. You should definitely take a look at this.

7.1 Commands

- `laplace(f(t), t, s)`

This generates the Laplace Transform of the function $f(t)$. The variable t must be explicitly defined as a variable earlier in the code (so you must have a line that says `t = var('t')` earlier in the code).

- `f.laplace(t,s)`

This is an alternate way to generate the Laplace Transform of $f(t)$. Both the variable t and the function f must be defined earlier in the code (so you must have things like `t = var('t')` and `f=sin(t)` earlier in the code).

- `inverse_laplace(g(s), s, t)`

This generates the Inverse Laplace Transform of the function $g(s)$. The variable s must be explicitly defined as a variable earlier in the code (so you must have a line that says `s = var('s')` earlier in the code).

- `g.inverse_laplace(s,t)`

This is an alternate way to generate the Inverse Laplace Transform of $g(s)$. Both the variable s and the function g must be defined earlier in the code (so you must have things like `s = var('s')` and `g=1/(s^2+1)` earlier in the code).

- `f.partial_fraction()`

This finds the partial fraction decomposition of the function f (defined in terms of some variable, usually s). Both the variable s and the function f must be defined earlier in the code (so you must have things like `s = var('s')` and `f=(s-1)/(s^2-4)` earlier in the code).

7.2 Example Code

- Here's some code that generates the Laplace Transform of a piece-wise defined function:

```
t, s = var('t s')
f = piecewise([(0,2),1],[(2, infinity),0])
f.laplace(t,s)
```

- Here's another example finding the Laplace Transform of $f(t) = t^2 e^t - \sin(t)$:

```
s, t = var('s t')
f = t^2*exp(t) - sin(t)
f.laplace(t,s)
```

- This is an example solving a system of equations using the Laplace Transform. It is a much more involved example:

```
s, t, X, Y = var('s t X Y')
x = function('x')
y = function('y')
de1 = 2*diff(x(t),t,2) + 6*x(t) - 2*y(t) == 0
de2 = diff(y(t),t, 2) + 2*y(t) - 2*x(t) == 0
sol1 = de1.laplace(t, s)
```

```

sol2 = de2.laplace(t, s)
s1 = str(sol1).replace("laplace(x(t), t, s)", "X").replace("laplace(y(t), t, s)", "Y")
s2 = str(sol2).replace("laplace(x(t), t, s)", "X").replace("laplace(y(t), t, s)", "Y")
s1ic = str(s1).replace("x(0)", "3").replace("D[0](x)(0)", "0")
s1ic = str(s1ic).replace("y(0)", "3").replace("D[0](y)(0)", "0")
s2ic = str(s2).replace("x(0)", "3").replace("D[0](x)(0)", "0")
s2ic = str(s2ic).replace("y(0)", "3").replace("D[0](y)(0)", "0")
LTX = sage_eval(s1ic, locals="s":s, "X":X, "Y":Y)
LTY = sage_eval(s2ic, locals="s":s, "X":X, "Y":Y)
eqns = [LTX, LTY]
s_sols = solve(eqns, X, Y)
ltx = s_sols[0][0].rhs()
lty = s_sols[0][1].rhs()
sols = [inverse_laplace(ltx, s, t), inverse_laplace(lty, s, t)]
sols

```

8 General Comments

- It's hard to overestimate how useful the Laplace Transform is. Pretty much any constant-coefficient linear equation that you can solve by other means is easier to solve using the Laplace Transform. Where it really comes into its own is with nonhomogeneous equations. These can be very labor-intensive to solve by other means. The Laplace Transform gives us a systematic way to attack these problems.
- One of the great benefits of using the Laplace Transform is that it allows you to solve IVPs in one step. Usually (meaning without using the Laplace Transform), you need to find the general solution to the given differential equation and then plug the initial conditions in to solve for the constants. This is a two-step process, the first involving techniques ideas from differential equations and the second involving more computational and algebraic work. With the Laplace Transform, these two steps are combined into one and it is usually more straight-forward to find the solution.
- If you have a nonhomogeneous equation with a forcing term that involves piece-wise defined functions, step functions, or delta functions, the Laplace Transform is pretty much the only game in town. These kinds of forcing functions occur frequently in engineering, especially in electrical engineering. If you are heading into those fields, you will want to become very familiar with the Laplace Transform.
- While we don't have the time to go into any of this in depth, the Laplace Transform also allows us to evaluate many complicated integrals, develop series expansions for functions, prove things about various special functions that arise naturally in mathematical physics, and many other analytical things that are not directly related to solving differential equations. It is an immensely powerful analytical tool. It also has the nice property of basically combining everything that you learned about integration in Calc 2 into a single conceptual framework. Why do we care about improper integrals? What practical applications does integration by parts have? When will we ever actually use partial fractions? The answer to all of these questions can be given in terms of the Laplace Transform.
- There is a complex-valued generalization of the Laplace Transform that is arguably the most important idea in applied analysis (perhaps even in all of applied mathematics). This is the celebrated **Fourier Transform**. This idea revolutionized mathematics in the 19th century when it was first introduced and is the computational foundation for pretty much all of modern technology. Any time you use anything that sends or receives signals of any sort (think radio, television, cellphones, the internet, etc) somewhere a Fourier Transform is being calculated.
- On a somewhat more abstract note, the Laplace Transform is in some sense the "simplest" **integral transform**. These are immensely important in most advanced branches of analysis. They are used everywhere and they all share certain properties with the Laplace Transform. If you want to dive

deeper into modern mathematics and its applications, integral transforms are unavoidable. The best way to get a sense of what they are, how they are used, and why they are important is to start with the most ‘basic’ example, which is the Laplace Transform.

- On top of all of this, the Laplace Transform is just an incredibly beautiful mathematical object. It translates problems that are formulated in terms of differential equations into problems that can be solved algebraically. This is a profound step, one that opens so many doors and leads to so many unexpected insights.

9 Reading Notes

- The Laplace Transform (J 6.1) (L 6.1)
- Transforms of IVPs (J 6.2) (L 6.2)
- Unit Step and Unit Impulse (Dirac Delta) Function (J Example 6.1.5, Example 6.1.7, 6.3) (L 6.4)
- Convolution (J 6.4) (L 6.3)

10 Additional Topics

Look in the indices of Judson and Lebl for these topics.

- Square and Triangular Wave Functions
- The Laplace Transform of a Periodic Function with Period T .
- Transfer Functions
- The Volterra Integral Equation

11 Homework Exercises

Please see the **Updated Schedule and Homework** document in the Google Drive to find the problems to turn in.

11.1 For Practice

- Lebl 6.1) 101, 102, 103, 104
- Lebl 6.2) 102, 103, 104
- Lebl 6.3) 101, 102, 103
- Lebl 6.4) 101, 102, 103, 104